DATABASE MANAGEMENT SYSTEM LAB

Sub Code: 18IS507
Credits : 01
Hrs. /Week: 0+0+2+0 **Total Hours : 36**

COURSE LEARNING OBJECTIVES

This Course will enable students to

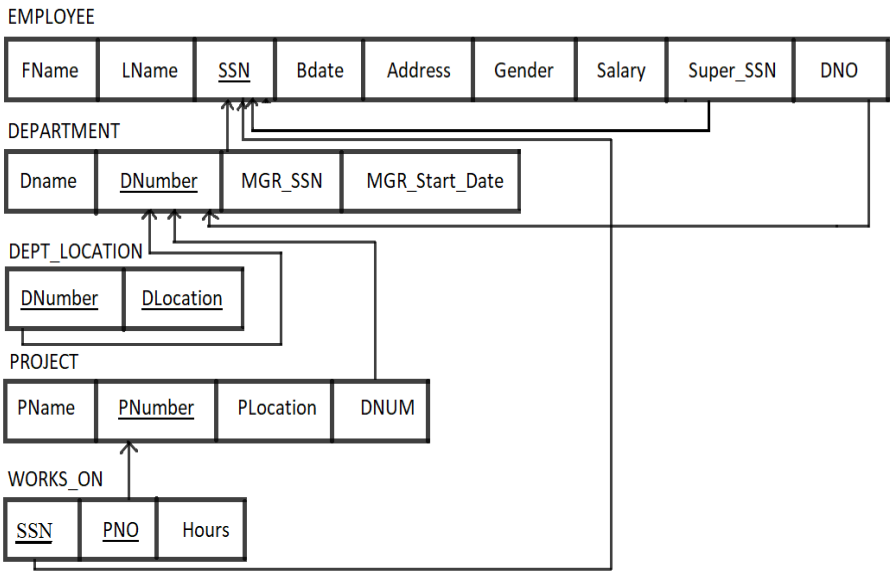
1. Learn how to design ER and schema diagrams for the given database problems and understand the mapping structure of entity relationship to tables.
2. Program SQL queries using Data Definition and Data Manipulation Languages
3. Program SQL queries through a variety of database problems.
4. Understand the concept of stored procedures, triggers and cursors in developing database applications.
5. Make use of complex and advanced query concepts in the developments of real time application using the database.

Course Content:

PART A:

1. Consider the schema for Employee Database:
EMPLOYEE(SSN, FName, LName, Address, Sex, Salary, SuperSSN, DNo)
DEPARTMENT(DNo, DName, MgrSSN, MgrStartDate)
DLOCATION(DNo,DLoc)
PROJECT(PNo, PName, PLocation, DNo)
WORKS_ON(SSN, PNo, Hours)
DEPENDENT(ESSN, Name, Sex, Bdate, Relationship)

SCHEMA DIAGRAM:



CREATE STATEMENTS:

EMPLOYEE table:

```
CREATE table employee
(
SSN varchar(10) primary key,
FName varchar(20) NOT NULL,
LName varchar(20),
Address varchar(20),
Gender CHAR(1) check (Gender in ('F','M')),
Salary decimal(10,2),
SuperSSN varchar(10) references Employee(SSN),
DNO int
);
```

DEPARTMENT table:

```
create table Department
(
DNUM int primary key check (DNUM > 0 and DNUM <= 20),
DName varchar(30) UNIQUE NOT NULL,
MGRSSN varchar(10) references Employee(SSN),
MGRStartDtae Date
);
```

DEPT_LOCATION table:

```
Create table DLocation
(
DNO int references Department(DNUM),
DLoc Varchar(30),
primary key (DNO, DLoc)
);
```

PROJECT table:

```
Create table Project
(
PNO int primary key,
PName varchar(30),
PLocation varchar(30),
DNO int references Department(DNUM)
);
```

WORKS_ON table:

```
Create table Works_on
(
ESSN varchar(10) references Employee(SSN),
PNO int references Project(PNO),
hours int,
primary key(ESSN, PNO)
);
```

ALTER STATEMENTS:

EMPLOYEE table:

```
alter table Employee add constraint FK_Emp_Dep FOREIGN KEY(DNO) references
Department(DNUM)
```

INSERT STATEMENTS:

EMPLOYEE table:

```
insert into Employee values('1111111111', 'Aishwarya', 'S', 'Tamil Nadu', 'F',
100000, NULL, NULL)
insert into Employee values('1111111112', 'Bhoomika', 'Bhat', 'Bangalore', 'F',
200000, NULL, NULL)
insert into Employee values('1111111113', 'Suhas', 'S', 'Andra Pradesh', 'M',
900000, NULL, NULL)
```

```
insert into Employee values('1111111116', 'Nidhi', 'Prabhu', 'Thirthahalli', 'F',
250000, NULL, NULL)
insert into Employee values('1111111114', 'Deeksha', 'D', 'Udupi', 'F', 800000,
NULL, NULL)
insert into Employee values('1111111115', 'Adithi', 'M', 'Manipal', 'F', 500000,
NULL, NULL)
```

DEPARTMENT table:

```
insert into Department values(1, 'Information Science', '1111111115', '07-10-2020')
insert into Department values(2, 'Electroncis and Communication', '1111111112',
'09-05-2015')
insert into Department values(3, 'Electrical and Electronics', '1111111111', '09-
09-2014')
insert into Department values(4,'Biotech','1111111114','12-15-1990')
```

DEPT_LOCATION table:

```
insert into DLocation values(1, 'Udupi')
insert into DLocation values(2, 'Bangalore')
insert into DLocation values(3, 'Mangalore')
insert into DLocation values(4, 'Manipal')
```

PROJECT table:

```
insert into Project values(20, 'P1', 'Mysore', 1)
insert into Project values(21, 'P2', 'Udupi', 3)
insert into Project values(22, 'P3', 'Hubli', 2)
insert into Project values(23, 'P4', 'Surathkal', 2)
insert into Project values(24, 'P5', 'Bangalore', 3)
```

WORKS_ON table:

```
insert into Works_on values('1111111111', '20', 7)
insert into Works_on values('1111111112', '21', 5)
insert into Works_on values('1111111113', '22', 6)
insert into Works_on values('1111111114', '23', 8)
insert into Works_on values('1111111115', '24', 8)
insert into Works_on values('1111111115', '20', 8)
insert into Works_on values('1111111116', '21', 5)
insert into Works_on values('1111111113', '14', 7)
```

UPDATE STATEMENTS:

EMPLOYEE table:

```
update Employee set SuperSSN='1111111115' where SSN='1111111114'
update Employee set SuperSSN='1111111114' where SSN='1111111111'
update Employee set SuperSSN='1111111114' where SSN='1111111112'
update Employee set SuperSSN='1111111114' where SSN='1111111113'
update Employee set SuperSSN='1111111114' where SSN='1111111116'

update Employee set DNO=1 where SSN='1111111111'
update Employee set DNO=2 where SSN='1111111112'
update Employee set DNO=3 where SSN='1111111113'
update Employee set DNO=1 where SSN='1111111114'
update Employee set DNO=2 where SSN='1111111115'
update Employee set DNO=1 where SSN='1111111116'

Update Employee set LName='Codd' where SSN=1111111115
Update Project set PName='Smart City' where PNO=24
Update Department set DName='Research' where DNUM='2'
```

SELECT * FROM <TABEL_NAME> for all tables:

```
Select * from Employee
```

	SSN	FName	LName	Address	Gender	Salary	SuperSSN	DNO
1	1111111111	Aishwarya	S	Tamil Nadu	F	100000.00	1111111114	1
2	1111111112	Bhoomika	Bhat	Bangalore	F	200000.00	1111111114	2
3	1111111113	Suhas	S	Andra Pradesh	M	900000.00	1111111114	3
4	1111111114	Deeksha	D	Udupi	F	800000.00	1111111115	1
5	1111111115	Adithi	M	Manipal	F	500000.00	NULL	2
6	1111111116	Nidhi	Prabhu	Thirthahalli	F	250000.00	1111111114	1

```
Select * from Department
```

	DNUM	DName	MGRSSN	MGRStartDtae
1	1	Information Science	1111111115	2020-07-10
2	2	Electroncis and Communication	1111111112	2015-09-05
3	3	Electrical and Electronics	1111111111	2014-09-09
4	4	Biotech	1111111114	1990-12-15

```
select * from DLocation
```

	DNO	DLoc
1	1	Udupi
2	2	Bangalore
3	3	Mangalore
4	4	Manipal

```
Select * from project
```

	PNO	PName	PLocation	DNO
1	20	P1	Mysore	1
2	21	P2	Udupi	3
3	22	P3	Hubli	2
4	23	P4	Surathkal	2
5	24	P5	Bangalore	3

```
Select * from Works_on
```

	ESSN	PNO	hours
1	1111111111	20	7
2	1111111112	21	5
3	1111111113	22	6
4	1111111114	23	8
5	1111111115	20	8
6	1111111115	24	8
7	1111111116	21	5

Write SQL queries to

1. Make a list of all project numbers for projects that involve an employee whose last name is 'Codd', either as a worker or as a manager of the department that controls the project.

   select distinct PNO from Works_on where ESSN=
   (select SSN from employee where LName='Codd')
   union
   select distinct PNO from Project as p, Department as d
   where p.DNO=d.DNUM AND d.MGRSSN=(select SSN from employee where LName = 'Codd')

2. Show the resulting salaries if every employee working on the 'Smart City' project is given a 10 percent raise.

   Update Project set PName='Smart City' where PNO=24

   select SSN, FName, LName, 1.1*Salary as Updated_salary

```
from employee where SSN in --cannot give = here as 2 ppl are working
(select ESSN from Works_on where PNO =
(select PNO from project where PName='Smart city'))
```

3. Find the sum of the salaries of all employees of the 'Information Science' department, as well as the maximum salary, the minimum salary, and the average salary in this department.

```
Update Department set DName='Research' where DNUM='2'

select SUM(e.salary) as Total_Salary, MAX(e.salary) as Max_Salary
, MIN(e.salary) as Min_Salary, AVG(e.salary) as Avg_Salary
from employee e , Department d
where d.DName='Research' and e.DNO=d.DNUM
```

4. Retrieve the name of each employee who works on all the projects controlled by department number 5.

```
select e.FName,e.LName
from Department d,employee e,Works_on w,Project p
where d.DNUM=e.DNO and e.SSN=w.ESSN and w.PNO=p.PNO and
p.DNO=3
```

5. For each department that has more than five employees, retrieve the department number and the number of its employees who are making more than Rs. 10,00,000.

```
select DNO, count(*) as Num_of_emps
from employee
where Salary > 100000 and DNO in
(select DNO
from employee group by DNO
having count(*)>2)
group by DNO
```

2. Consider the Insurance database given below.

PERSON (**driver – id #:** String, name: string, address: string)
CAR (**regno**: string, model: string, year: int)
ACCIDENT (**report-number**: int, **accd-date:** date, location: string)
OWNS (**driver-id #:** string, **regno**: string)
PARTICIPATED (**driver-id:** string, **Regno:** string, **report-number:** int, damage amount: int)

SCHEMA DIAGRAM:

CREATE STATEMENTS:

PERSON table:

```
CREATE TABLE PERSON
(
D_ID VARCHAR(10) PRIMARY KEY,
D_NAME VARCHAR(25) NOT NULL,
ADDR VARCHAR(30)
);
```

CAR table:

```
CREATE TABLE CAR
(
REG_NO VARCHAR(10) PRIMARY KEY,
MODEL VARCHAR(10) NOT NULL,
YEAR DATE
);
```

ACCIDENT table:

```
CREATE TABLE ACCIDENT
(
REPORT_NUM INT PRIMARY KEY,
ACC_DATE DATE,
ACC_LOC VARCHAR(30)
);
```

OWNS table:

```
CREATE TABLE OWNS
(
D_ID VARCHAR(10) REFERENCES PERSON(D_ID),
REG_NO VARCHAR(10) REFERENCES CAR(REG_NO),
PRIMARY KEY(D_ID,REG_NO)
);
```

PARTICIPATED table:

```
CREATE TABLE PARTICIPATED
(
D_ID VARCHAR(10) REFERENCES PERSON(D_ID),
```

```
REG_NO VARCHAR(10) REFERENCES CAR(REG_NO),
REPORT_NUM INT REFERENCES ACCIDENT(REPORT_NUM),
DAM_AMOUNT INT,
PRIMARY KEY (D_ID,REG_NO,REPORT_NUM)
);
```

INSERT STATEMENTS:
AND
SELECT * FROM <TABEL_NAME> for all tables:

PERSON table:

```
INSERT INTO PERSON VALUES('00001','AAAA','UDUPI')
INSERT INTO PERSON VALUES('00002','BBBB','MANGALORE')
INSERT INTO PERSON VALUES('00003','CCCC','BANGALORE')
INSERT INTO PERSON VALUES('00004','EEEE','MANGALORE')
INSERT INTO PERSON VALUES('00005','FFFF','BANGALORE')
INSERT INTO PERSON VALUES('00006','GGGG','MANGALORE')

SELECT * FROM PERSON
```

	D_ID	D_NAME	ADDR
1	00001	AAAA	UDUPI
2	00002	BBBB	MANGALORE
3	00003	CCCC	BANGALORE
4	00004	EEEE	MANGALORE
5	00005	FFFF	BANGALORE
6	00006	GGGG	MANGALORE

CAR table:

```
INSERT INTO CAR VALUES('101','A','2000-03-28')
INSERT INTO CAR VALUES('102','B','2000-04-28')
INSERT INTO CAR VALUES('103','C','2000-05-28')
INSERT INTO CAR VALUES('104','D','2000-06-28')
INSERT INTO CAR VALUES('105','E','2000-07-28')
INSERT INTO CAR VALUES('106','F','2000-08-28')

SELECT * FROM CAR
```

	REG_NO	MODEL	YEAR
1	101	A	2000-03-28
2	102	A	2000-04-28
3	103	C	2000-05-28
4	104	D	2000-06-28
5	105	E	2000-07-28
6	106	F	2000-08-28
7	107	A	2000-02-28

ACCIDENT table:

```
INSERT INTO ACCIDENT VALUES(1,'2002-04-25','UDUPI')
INSERT INTO ACCIDENT VALUES(2,'2003-05-05','BANGALORE')
INSERT INTO ACCIDENT VALUES(3,'2005-06-26','MANGALORE')
INSERT INTO ACCIDENT VALUES(4,'2002-07-15','UDUPI')
INSERT INTO ACCIDENT VALUES(5,'2008-08-03','MANGALORE')

SELECT * FROM ACCIDENT
```

	REPORT_NUM	ACC_DATE	ACC_LOC
1	1	2002-04-25	UDUPI
2	2	2003-05-05	BANGALORE
3	3	2005-06-26	MANGALORE
4	4	2002-07-15	UDUPI
5	5	2008-08-03	MANGALORE

OWNS table:

```
INSERT INTO OWNS VALUES('00001','101')
INSERT INTO OWNS VALUES('00002','102')
INSERT INTO OWNS VALUES('00003','103')
INSERT INTO OWNS VALUES('00004','104')
INSERT INTO OWNS VALUES('00005','105')
INSERT INTO OWNS VALUES('00001','106')

UPDATE OWNS SET D_ID='00006' WHERE REG_NO='106'

SELECT * FROM OWNS
```

	D_ID	REG_NO
1	00001	101
2	00001	107
3	00002	102
4	00003	103
5	00004	104
6	00005	105
7	00006	106

PARTICIPATED table:

```
INSERT INTO PARTICIPATED VALUES('00001','101',1,2000)
INSERT INTO PARTICIPATED VALUES('00002','102',2,2000)
INSERT INTO PARTICIPATED VALUES('00003','103',3,2000)
INSERT INTO PARTICIPATED VALUES('00004','104',4,2000)
INSERT INTO PARTICIPATED VALUES('00005','105',5,2000)
INSERT INTO PARTICIPATED VALUES('00006','106',5,2000)

UPDATE PARTICIPATED SET DAM_AMOUNT=5000 WHERE REG_NO='106'
UPDATE PARTICIPATED SET DAM_AMOUNT=6000 WHERE REG_NO='104'
UPDATE PARTICIPATED SET DAM_AMOUNT=3000 WHERE REG_NO='103'
UPDATE PARTICIPATED SET DAM_AMOUNT=4000 WHERE REG_NO='102'

SELECT * FROM PARTICIPATED
```

	D_ID	REG_NO	REPORT_NUM	DAM_AMOUNT
1	00001	101	1	2000
2	00002	102	2	4000
3	00003	103	3	3000
4	00004	104	4	6000
5	00005	105	5	2000
6	00006	106	5	5000

Write SQL queries to

1. List the names of people who owned cars that were involved in accidents in 2008.

   SELECT P.D_NAME
   FROM PERSON P,PARTICIPATED P1,ACCIDENT A
   WHERE P.D_ID =P1.D_ID AND P1.REPORT_NUM=A.REPORT_NUM AND ACC_DATE LIKE '2008-__-__'

2. Find the name of owner and his car that has maximum number of accidents in 2008

   SELECT P.D_NAME,C.MODEL
   FROM PERSON P,CAR C,ACCIDENT A,PARTICIPATED P1
   WHERE P.D_ID=P1.D_ID AND C.REG_NO=P1.REG_NO AND A.REPORT_NUM=P1.REPORT_NUM AND A.ACC_DATE LIKE '2008-__-__'
   GROUP BY P.D_NAME,C.MODEL
   HAVING COUNT(*)>= ALL(SELECT COUNT(*)
   FROM PERSON P2,CAR C1,ACCIDENT A1,PARTICIPATED P4

```
WHERE P2.D_ID=P4.D_ID AND C1.REG_NO=P4.REG_NO AND A1.REPORT_NUM=P4.REPORT_NUM AND A1.ACC_DATE LIKE '2008-__-__'
GROUP BY P2.D_NAME,C1.MODEL)
```

3. List the name of owners who own atleast two TOYOTA cars.

```
SELECT P.D_NAME
FROM PERSON P, CAR C,OWNS O
WHERE P.D_ID=O.D_ID AND C.REG_NO=O.REG_NO AND C.MODEL='A'
GROUP BY P.D_NAME
HAVING COUNT(*)>=1

UPDATE CAR SET MODEL='A' WHERE REG_NO=102
```

4. List the name of owner who owns maximum TOYOTA cars.

```
INSERT INTO CAR VALUES ('107','A','2000-02-28')

INSERT INTO OWNS VALUES('00001','107')

SELECT P.D_NAME,COUNT(*) AS NUM_OF_CARS
FROM PERSON P,CAR C,OWNS O
WHERE P.D_ID=O.D_ID AND O.REG_NO=C.REG_NO AND C.MODEL='A'
GROUP BY P.D_NAME
HAVING COUNT(*) >= ALL
(SELECT COUNT(*) AS NUM_OF_CARS
FROM PERSON P1,CAR C1,OWNS O1
WHERE P1.D_ID=O1.D_ID AND O1.REG_NO=C1.REG_NO AND C1.MODEL='A'
GROUP BY P1.D_NAME)
```

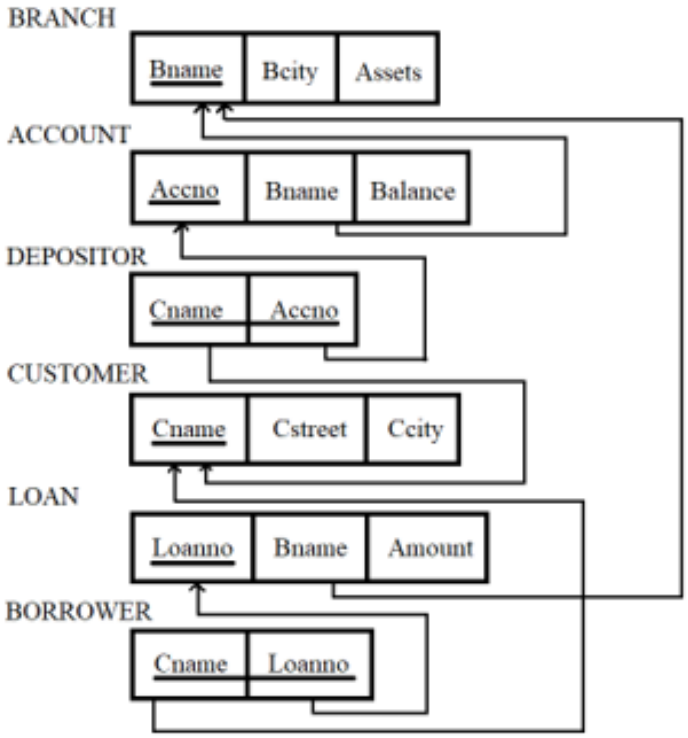
5. Find the name of owner who owns cars having minimum damage amount for accidents in 2008

```
SELECT P.D_NAME,SUM(P1.DAM_AMOUNT)
FROM PERSON P,ACCIDENT A,PARTICIPATED P1
WHERE P.D_ID=P1.D_ID AND A.REPORT_NUM=P1.REPORT_NUM AND A.ACC_DATE LIKE '2008-__-__'
GROUP BY P.D_NAME
HAVING SUM(P1.DAM_AMOUNT) <= ALL
(SELECT SUM(P4.DAM_AMOUNT)
FROM PERSON P2,ACCIDENT A1,PARTICIPATED P4
```

3. Consider the following database for a banking enterprise:

BRANCH (**branch-name**: string, branch-city: string, assets: real)
ACCOUNT (**accno:** int, branch-name: string, balance: real)
DEPOSITOR (**customer-name**: string, **accno**: int)
CUSTOMER (**customer-name**: string, customer-street: string, customer-city: string)
LOAN (**loan-number**: int, branch-name: string, amount: real)
BORROWER (**customer-name**: string, **loan-number:** int)

SCHEMA DIAGRAM:



Write SQL queries to

1. Find all the customers who have at least two accounts at the Main branch. (*)
2. Find all the customers who have an account at all the branches located in a specific city (*)
3. Find all the customers who have accounts in at least 2 branches located in a specific city.
4. Find all the customers who have borrowed loan from at least one branch that is located in a specific city.
5. Find the branch name that has maximum number of customers in a specific city

CREATE STATEMENTS:

BRANCH TABLE:

```
create table Branch(
bname varchar(15),
bcity varchar(15),
assets real,
primary key(bname)
);
```

ACCOUNT TABLE:

```
create table Account(
accno int,
bname varchar(15) default 'Manipal',
balance real,
primary key(accno),
foreign key(bname) references Branch(bname) on delete cascade on update cascade
);
```

CUSTOMER TABLE:

```
create table Customer
(
cname varchar(20),
cstreet varchar(25),
ccity varchar(20),
primary key(cname),
```

```
);
```

LOAN TABLE:

```
create table Loan
(loan_no int,
bname varchar(15),
amount real,
primary key(loan_no),
foreign key(bname)references Branch(bname) on delete cascade on update cascade
);
```

BORROWER TABLE:

```
create table Borrower
(cname varchar(20),
loan_no int,
primary key(cname,loan_no),
foreign key(cname)references Customer(cname) on delete cascade on update cascade,
foreign key(loan_no)references Loan(loan_no) on delete cascade on update cascade,
unique(loan_no)
);
```

DEPOSITOR TABLE:

```
create table Depositor
(
cname varchar(20),
accno int,
primary key(cname,accno),
foreign key(accno)references Account(accno) on delete cascade on update cascade,
foreign key(cname)references Customer(cname) on delete cascade on update cascade,
unique(accno)
);
```

INSERT STATEMENTS:

BRANCH TABLE:

```
insert into Branch values('SBI_Udupi','Udupi','30000')
insert into Branch values('SBI_Mangalore','Mangalore','20000')
insert into Branch values('SBI_Manipal','Udupi','40000')
insert into Branch values('SBI_Karkala','Karkala','30000')
insert into Branch values('SBI_Bangalore','Bangalore','20000')
insert into Branch values('SBI_Sasthana','Udupi','40000')
insert into Branch values('SBI_Surathkal','Mangalore','40000')
```

ACCOUNT TABLE:

```
insert into Account values('123456781','SBI_Udupi','300000')
insert into Account values('123456782','SBI_Bangalore','310000')
insert into Account values('123456783','SBI_Sasthana','200000')
insert into Account values('123456784','SBI_Mangalore','500000')
insert into Account values('123456785','SBI_Surathkal','360000')
insert into Account values('123456786','SBI_Udupi','870000')
insert into Account values('123456787','SBI_Bangalore','650000')
insert into Account values('123456788','SBI_Manipal','40000')
insert into Account values('123456789','SBI_Mangalore','230000')
insert into Account values('123456770','SBI_Karkala','350000')
insert into Account values('123456771','SBI_Manipal','130000')
insert into Account values('123456772','SBI_Bangalore','350000')
insert into Account values('123456773','SBI_Bangalore','360000')
insert into Account values('123456774','SBI_Mangalore','700000')
insert into Account values('123456775','SBI_Udupi','800000')
```

```sql
insert into Account values('123456791','SBI_Bangalore','360000')
insert into Account values('123456792','SBI_Bangalore','380000')
insert into Account values('123456793','SBI_Sasthana','400000')
insert into Account values('123456794','SBI_Surathkal','600000')
insert into Account values('123456795','SBI_Udupi','360000')
```

CUSTOMER TABLE:

```sql
insert into Customer values('Nidhi','8th cross','Udupi');
insert into Customer values('Maneesha','4th cross','Mangalore');
insert into Customer values('Mallik','2nd cross','Karkala');
insert into Customer values('Nisha','5th cross','Bangalore');
insert into Customer values('Sanjana','13th cross','Mangalore');
insert into Customer values('Rahul','7th cross','Udupi');
```

DEPOSITOR TABLE:

```sql
insert into Depositor values('Rahul','123456781')
insert into Depositor values('Mallik','123456785')
insert into Depositor values('Nidhi','123456784')
insert into Depositor values('Sanjana','123456786')
insert into Depositor values('Nidhi','123456783')
insert into Depositor values('Maneesha','123456782')
insert into Depositor values('Sanjana','123456787')
insert into Depositor values('Mallik','123456788')
insert into Depositor values('Nidhi','123456789')
insert into Depositor values('Sanjana','123456771')
insert into Depositor values('Mallik','123456773')
insert into Depositor values('Maneesha','123456772')
insert into Depositor values('Rahul','123456770')
insert into Depositor values('Mallik','123456774')
insert into Depositor values('Nidhi','123456791')
insert into Depositor values('Sanjana','123456775')
insert into Depositor values('Nisha','123456792')
insert into Depositor values('Maneesha','123456793')
insert into Depositor values('Mallik','123456794')
insert into Depositor values('Maneesha','123456795')
```

LOAN TABLE:

```sql
insert into Loan values(1,'SBI_Karkala','12000');
insert into Loan values(2,'SBI_Udupi','15000');
insert into Loan values(3,'SBI_Manipal','20000');
insert into Loan values(4,'SBI_Bangalore','30000');
insert into Loan values(5,'SBI_Mangalore','13000');
insert into Loan values(6,'SBI_Surathkal','13000');
insert into Loan values(7,'SBI_Sasthana','13000');
insert into Loan values(8,'SBI_Manipal','25000');
insert into Loan values(9,'SBI_Bangalore','35000');
insert into Loan values(10,'SBI_Udupi','65000');
insert into Loan values(11,'SBI_Mangalore','33000');
insert into Loan values(12,'SBI_Karkala','62000');
insert into Loan values(13,'SBI_Udupi','35000');
insert into Loan values(14,'SBI_Manipal','17000');
```

BORROWER TABLE:

```sql
insert into Borrower values('Nidhi',9);
insert into Borrower values('Mallik',6);
insert into Borrower values('Maneesha',2);
insert into Borrower values('Nisha',4);
insert into Borrower values('Sanjana',12);
insert into Borrower values('Rahul',3);
insert into Borrower values('Nisha',14);
insert into Borrower values('Nidhi',5);
insert into Borrower values('Mallik',7);
insert into Borrower values('Maneesha',8);
```

```
insert into Borrower values('Sanjana',11);
insert into Borrower values('Rahul',13);
insert into Borrower values('Nisha',10);
insert into Borrower values('Sanjana',1);
```

SELECT * FROM <TABEL_NAME> for all tables:

```
select * from Branch
```

	bname	bcity	assets
1	SBI_Bangalore	Bangalore	20000
2	SBI_Karkala	Karkala	30000
3	SBI_Mangalore	Mangalore	20000
4	SBI_Manipal	Udupi	40000
5	SBI_Sasthana	Udupi	40000
6	SBI_Surathkal	Mangalore	40000
7	SBI_Udupi	Udupi	30000

```
select * from Account
```

	accno	bname	balance
1	123456770	SBI_Karkala	350000
2	123456771	SBI_Manipal	130000
3	123456772	SBI_Bangalore	350000
4	123456773	SBI_Bangalore	360000
5	123456774	SBI_Mangalore	700000
6	123456775	SBI_Udupi	800000
7	123456781	SBI_Udupi	300000
8	123456782	SBI_Bangalore	310000
9	123456783	SBI_Sasthana	200000
10	123456784	SBI_Mangalore	500000
11	123456785	SBI_Surathkal	360000
12	123456786	SBI_Udupi	870000
13	123456787	SBI_Bangalore	650000
14	123456788	SBI_Manipal	40000
15	123456789	SBI_Mangalore	230000
16	123456791	SBI_Bangalore	360000
17	123456792	SBI_Bangalore	380000
18	123456793	SBI_Sasthana	400000
19	123456794	SBI_Surathkal	600000
20	123456795	SBI_Udupi	360000

```
select * from Customer
```

	cname	cstreet	ccity
1	Mallik	2nd cross	Karkala
2	Maneesha	4th cross	Mangalore
3	Nidhi	8th cross	Udupi
4	Nisha	5th cross	Bangalore
5	Rahul	7th cross	Udupi
6	Sanjana	13th cross	Mangalore

```
select * from Depositor
```

	cname	accno
1	Rahul	123456770
2	Sanjana	123456771
3	Maneesha	123456772
4	Mallik	123456773
5	Mallik	123456774
6	Sanjana	123456775
7	Rahul	123456781
8	Maneesha	123456782
9	Nidhi	123456783
10	Nidhi	123456784
11	Mallik	123456785
12	Sanjana	123456786
13	Sanjana	123456787
14	Mallik	123456788
15	Nidhi	123456789
16	Nidhi	123456791
17	Nisha	123456792
18	Maneesha	123456793
19	Mallik	123456794
20	Maneesha	123456795

```
select * from Loan
```

	loan_no	bname	amount
1	1	SBI_Karkala	12000
2	2	SBI_Udupi	15000
3	3	SBI_Manipal	20000
4	4	SBI_Bangalore	30000
5	5	SBI_Mangalore	13000
6	6	SBI_Surathkal	13000
7	7	SBI_Sasthana	13000
8	8	SBI_Manipal	25000
9	9	SBI_Bangalore	35000
10	10	SBI_Udupi	65000
11	11	SBI_Mangalore	33000
12	12	SBI_Karkala	62000
13	13	SBI_Udupi	35000
14	14	SBI_Manipal	17000

```
select * from Borrower
```

	cname	loan_no
1	Sanjana	1
2	Maneesha	2
3	Rahul	3
4	Nisha	4
5	Nidhi	5
6	Mallik	6
7	Mallik	7
8	Maneesha	8
9	Nidhi	9
10	Nisha	10
11	Sanjana	11
12	Sanjana	12
13	Rahul	13
14	Nisha	14

1. Find all the customers who have at least two accounts at the Main branch. (*)

```
select d.cname,count(*) as No_of_accounts
from Account a, Depositor d
where a.accno=d.accno and a.bname='SBI_Udupi'
group by d.cname
```

```
having count(*)>1
```

2. Find all the customers who have an account at all the branches located in a specific city (*)

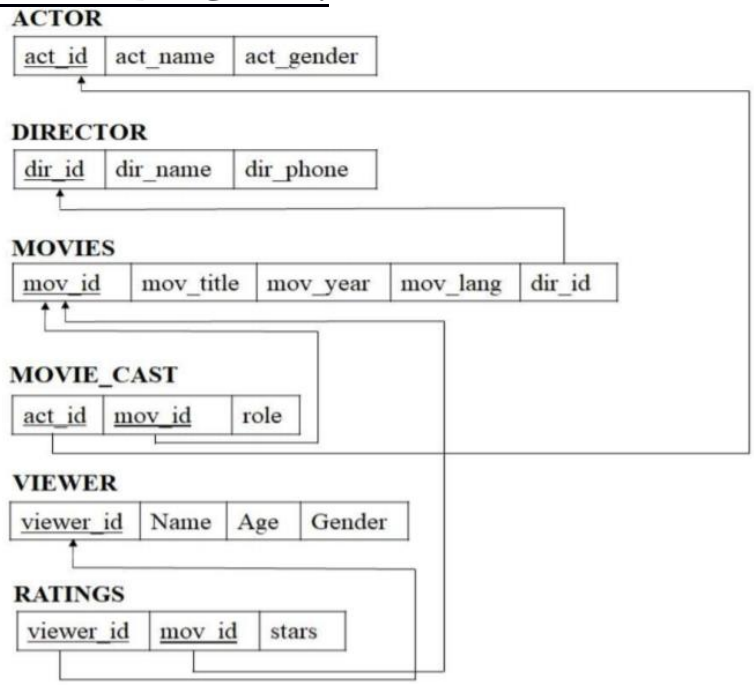
```
select *
from Customer C
where NOT EXISTS (
select B.bname
from Branch B
where B.bcity='Bangalore'

except

select distinct A.bname
from Depositor D, Account A, Branch B
where D.accno=A.accno AND A.bname=B.bname AND C.ccity='Bangalore'
AND D.cname=C.cname);
```

3. Find all the customers who have accounts in at least 2 branches located in a specific city.

```
select d.cname, count(distinct a.bname) number_of_branches
from Account a, Depositor d, Branch b
where a.accno=d.accno and a.bname=b.bname and b.bcity='Udupi'
group by d.cname
having count(distinct a.bname)>=2
```

4. Consider the schema for Movie Database:
ACTOR(**Act_id**, Act_Name, Act_Gender)
DIRECTOR(**Dir_id**, Dir_Name, Dir_Phone)
MOVIES(**Mov_id**, Mov_Title, Mov_Year, Mov_Lang, Dir_id)
MOVIE_CAST(**Act_id, Mov_id**, Role)
VIEWER(**Viewer_ID**, Name, Age, Gender)
RATINGS(**Viewer_ID, Mov_id** , Stars)

SCHEMA DIAGRAM:



CREATE STATEMENTS:

ACTOR table:

```
CREATE TABLE ACTOR
(
ACT_ID INT PRIMARY KEY,
ACT_NAME varchar(30),
ACT_GENDER CHAR(1) CHECK (ACT_GENDER IN('M','F'))
);
```

DIRECTOR table:

```
CREATE TABLE DIRECTOR
(
DIR_ID INT PRIMARY KEY,
DIR_NAME VARCHAR(25),
dir_phone INT
);
```

MOVIES table:

```
CREATE TABLE MOVIES
(
MOV_ID INT PRIMARY KEY,
MOV_TITLE VARCHAR(50),
MOV_YEAR INT,
MOV_LANG VARCHAR(20),
DIR_ID INT REFERENCES DIRECTOR(DIR_ID),
UNIQUE(MOV_TITLE)
);
```

MOVIE_CAST table:

```
CREATE TABLE MOVIE_CAST(
ACT_ID INT REFERENCES ACTOR(ACT_ID),
MOV_ID INT REFERENCES MOVIES(MOV_ID),
ROLE VARCHAR(20),
PRIMARY KEY(ACT_ID,MOV_ID)
);
```

VIEWER table:

```
CREATE TABLE VIEWER(
VIEW_ID INT PRIMARY KEY,
VIEW_NAME VARCHAR(25),
AGE INT,
VIEW_GENDER char(1) CHECK (VIEW_GENDER IN('F','M'))
);
```

RATINGS table:

```
CREATE TABLE RATINGS(
VIEW_ID INT REFERENCES VIEWER(VIEW_ID),
MOV_ID INT REFERENCES MOVIES(MOV_ID),
STARS INT CHECK(STARS>=0 AND STARS<=5)
PRIMARY KEY(VIEW_ID,MOV_ID),
);
```

INSERT STATEMENTS:
AND
SELECT * FROM <TABEL_NAME> for all tables:

ACTOR table:

```
INSERT INTO ACTOR VALUES (101,'SHAHID KAPOOR','M')
INSERT INTO ACTOR VALUES (102,'ANUSHKA SHARMA','F')
```

```
INSERT INTO ACTOR VALUES (103,'RANBIR KAPOOR','M')
INSERT INTO ACTOR VALUES (104,'ALIA BHAT','F')
INSERT INTO ACTOR VALUES (105,'PARINITHI CHOPRA','F')
INSERT INTO ACTOR VALUES (106,'VARUN DHAVAN','M')

SELECT *FROM ACTOR
```

	ACT_ID	ACT_NAME	ACT_GENDER
1	101	SHAHID KAPOOR	M
2	102	ANUSHKA SHARMA	F
3	103	RANBIR KAPOOR	M
4	104	ALIA BHAT	F
5	105	PARINITHI CHOPRA	F
6	106	VARUN DHAVAN	M

DIRECTOR table:

```
INSERT INTO DIRECTOR VALUES (1001,'AAAA',93876259)
INSERT INTO DIRECTOR VALUES (1002,'BBBB',83761947)
INSERT INTO DIRECTOR VALUES (1003,'CCCC',85679026)
INSERT INTO DIRECTOR VALUES (1004,'DDDD',97254689)
INSERT INTO DIRECTOR VALUES (1005,'EEEE',84732648)

SELECT *FROM DIRECTOR
```

	DIR_ID	DIR_NAME	dir_phone
1	1001	AAAA	93876259
2	1002	BBBB	83761947
3	1003	CCCC	85679026
4	1004	DDDD	97254689
5	1005	EEEE	84732648

MOVIES table:

```
INSERT INTO MOVIES VALUES (1,'JAB WE MET',2007,'HINDI',1001)
INSERT INTO MOVIES VALUES (2,'PK',2014,'ENGLISH',1004)
INSERT INTO MOVIES VALUES (3,'YEH JAWANI HAI DIWANI',2013,'KANNADA',1005)
INSERT INTO MOVIES VALUES (4,'STUDENT OF THE YEAR',2012,'HINDI',1003)
INSERT INTO MOVIES VALUES (5,'ISHAQZAADE',2012,'HINDI',1001)
INSERT INTO MOVIES VALUES (6,'ABCD2',2015,'ENGLISH',1002)
INSERT INTO MOVIES VALUES (7,'KABIR SINGH',2018,'HINDI',1001)

SELECT *FROM MOVIES
```

	MOV_ID	MOV_TITLE	MOV_YEAR	MOV_LANG	DIR_ID
1	1	JAB WE MET	2007	HINDI	1001
2	2	PK	2014	ENGLISH	1004
3	3	YEH JAWANI HAI DIWANI	2013	KANNADA	1005
4	4	STUDENT OF THE YEAR	2012	HINDI	1003
5	5	ISHAQZAADE	2012	HINDI	1001
6	6	ABCD2	2015	ENGLISH	1002
7	7	KABIR SINGH	2018	HINDI	1001

MOVIE_CAST table:

```
INSERT INTO MOVIE_CAST VALUES (101,1,'HERO')
INSERT INTO MOVIE_CAST VALUES (102,2,'HEROINE')
INSERT INTO MOVIE_CAST VALUES (103,3,'HERO')
INSERT INTO MOVIE_CAST VALUES (104,4,'HEROINE')
INSERT INTO MOVIE_CAST VALUES (105,5,'HEROINE')
INSERT INTO MOVIE_CAST VALUES (106,6,'HERO')
INSERT INTO MOVIE_CAST VALUES (101,7,'HERO')

SELECT *FROM MOVIE_CAST
```

	ACT_ID	MOV_ID	ROLE
1	101	1	HERO
2	101	7	HERO
3	102	2	HEROINE
4	103	3	HERO
5	104	4	HEROINE
6	105	5	HEROINE
7	106	6	HERO

VIEWER table:

```
INSERT INTO VIEWER VALUES (1,'ZZZZ',22,'M')
INSERT INTO VIEWER VALUES (2,'YYYY',24,'F')
INSERT INTO VIEWER VALUES (3,'XXXX',24,'M')
INSERT INTO VIEWER VALUES (4,'WWWW',38,'M')
INSERT INTO VIEWER VALUES (5,'NNNN',20,'F')
INSERT INTO VIEWER VALUES (6,'SSSS',20,'M')
INSERT INTO VIEWER VALUES (7,'UUUU',20,'F')
INSERT INTO VIEWER VALUES (8,'VVVV',20,'M')
```

```
SELECT *FROM VIEWER
```

	VIEW_ID	MOV_ID	STARS
1	1	3	5
2	1	5	4
3	2	6	5
4	3	4	4
5	3	5	4
6	4	2	4
7	5	1	5
8	6	1	5

RATINGS table:

```
INSERT INTO RATINGS VALUES (1,3,5)
INSERT INTO RATINGS VALUES (1,5,4)
INSERT INTO RATINGS VALUES (2,6,5)
INSERT INTO RATINGS VALUES (3,4,4)
INSERT INTO RATINGS VALUES (3,5,4)
INSERT INTO RATINGS VALUES (4,2,4)
INSERT INTO RATINGS VALUES (5,1,5)
INSERT INTO RATINGS VALUES (6,1,5)
```

```
SELECT *FROM RATINGS
```

	VIEW_ID	MOV_ID	STARS
1	1	3	5
2	1	5	4
3	2	6	5
4	3	4	4
5	3	5	4
6	4	2	4
7	5	1	5
8	6	1	5

Write SQL queries to

1. List all actors who acted in a movie before 2000 and also in a movie after 2017.

   SELECT A.ACT_NAME

   FROM ACTOR A,MOVIE_CAST MC,MOVIES M

```
WHERE A.ACT_ID=MC.ACT_ID AND MC.MOV_ID=M.MOV_ID AND MOV_YEAR<2012
INTERSECT
SELECT A.ACT_NAME
FROM ACTOR A,MOVIE_CAST MC,MOVIES M
WHERE A.ACT_ID=MC.ACT_ID AND MC.MOV_ID=M.MOV_ID AND MOV_YEAR>2017
```

2. Find the title of movies and number of stars for each movie that has at least one rating and find the highest number of stars that movie received. Sort the result by movie title.

```
UPDATE RATINGS SET STARS=5
WHERE MOV_ID IN(SELECT M.MOV_ID
                FROM MOVIES M,DIRECTOR D
                WHERE M.DIR_ID=D.DIR_ID AND D.DIR_NAME='BBBB')

SELECT D.DIR_NAME,R.STARS
FROM RATINGS R,MOVIES M,DIRECTOR D
WHERE STARS = 5 AND R.MOV_ID=M.MOV_ID AND D.DIR_ID=M.DIR_ID AND D.DIR_NAME='BBBB'
```

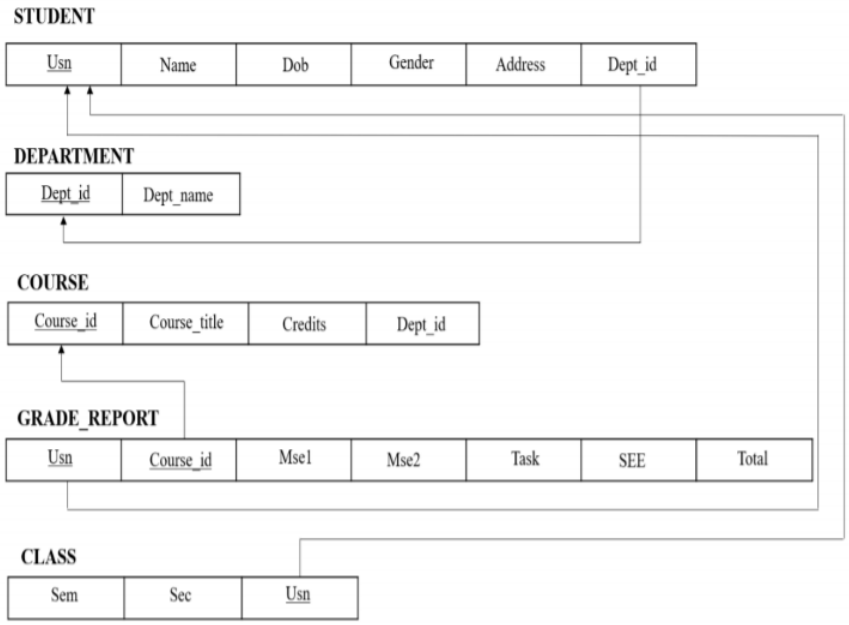
3. Update rating of all movies directed by 'Subhash Ghai' to 5.

```
SELECT M.MOV_TITLE,MAX(R.STARS) AS RATINGS
FROM MOVIES M,RATINGS R
WHERE M.MOV_ID=R.MOV_ID AND R.STARS>1
GROUP BY M.MOV_TITLE
```

5. Consider the schema for StudentDatabase:

STUDENT(**USN**, Name, DOB, Gender, Address,Dept_Id)
DEPARTMENT (**Dept_ID**, Dept_Name)
COURSE(**Course_Id**, Course_Title, Credits,Dept_Id)
GRADE_REPORT(**USN, Course_ID,** MSE1,MSE2,Task,SEE, Total)
CLASS(**Sem, Sec , USN**)

SCHEMA DIAGRAM:



CREATE STATEMENTS:

STUDENT table:

```
create table Student
(
usn varchar(10) primary key,
s_name varchar(15) not null,
dept_id int references Department(dept_id),
dob date,
gender char(1)
)
```

DEPARTMENT table:

```
create table Department
(
dept_id int primary key,
dept_name varchar(20) not null
)
```

COURSE table:

```
create table Course
(
course_id varchar(10) primary key,
course_title varchar(20) not null,
credits int,
dept_id int references Department(dept_id)
)
```

GRADE_REPORT table:

```
create table Grade_report
(
usn varchar(10) references Student(usn),
course_id varchar(10) references Course(course_id),
mse1 int,
mse2 int,
task int,
see int,
total int,
primary key (USN,course_id)
)
```

CLASS table:

```
create table Class
(
usn varchar(10) references Student(usn),
sem int not null,
sec char(1) not null,
primary key(usn,sem,sec)
)
```

INSERT STATEMENTS:
AND
SELECT * FROM <TABEL_NAME> for all tables:

STUDENT table:

```
insert into Student values('123456789','aaaaa',3,'2000-12-12','f')
insert into Student values('123456788','bbbbb',2,'2000-11-12','f')
insert into Student values('123456787','ccccc',4,'2000-10-12','m')
insert into Student values('123456786','ddddd',1,'2000-12-11','m')
insert into Student values('123456785','eeeee',3,'2000-12-10','f')
insert into Student values('123456784','fffff',2,'2000-9-12','m')
insert into Student values('123456783','ggggg',3,'2000-12-12','f')
insert into Student values('123456782','hhhhh',2,'2000-12-7','m')

select * from Student
```

	usn	s_name	dept_id	dob	gender
1	123456782	hhhhh	2	2000-12-07	m
2	123456783	ggggg	3	2000-12-12	f
3	123456784	fffff	2	2000-09-12	m
4	123456785	eeeee	3	2000-12-10	f
5	123456786	ddddd	1	2000-12-11	m
6	123456787	ccccc	4	2000-10-12	m
7	123456788	bbbbb	2	2000-11-12	f
8	123456789	aaaaa	3	2000-12-12	f

DEPARTMENT table:

```
insert into Department values (1,'ECE')
insert into Department values (2,'CSE')
insert into Department values (3,'ISE')
insert into Department values (4,'BT')

select * from Department
```

	dept_id	dept_name
1	1	ECE
2	2	CSE
3	3	ISE
4	4	BT

COURSE table:

```
insert into Course values ('ISE101','DS',4,1)
insert into Course values ('ECE101','DSD',3,1)
insert into Course values ('CSE101','DC',2,1)
insert into Course values ('ECE102','MPI',3,1)
insert into Course values ('ISE102','DSM',2,1)
insert into Course values ('CSE102','DAA',4,1)
insert into Course values ('BT101','MB',3,1)
insert into Course values ('BT102','MT',3,1)

select * from Course
```

	course_id	course_title	credits	dept_id
1	BT101	MB	3	1
2	BT102	MT	3	1
3	CSE101	DC	2	1
4	CSE102	DAA	4	1
5	ECE101	DSD	3	1
6	ECE102	MPI	3	1
7	ISE101	DS	4	1
8	ISE102	DSM	2	1

GRADE_REPORT table:

```
insert into Grade_report values('123456789','ISE101',20,19,7,47,93)
insert into Grade_report values('123456789','ISE102',20,19,7,48,94)

insert into Grade_report values('123456788','CSE101',20,19,9,47,95)
insert into Grade_report values('123456788','CSE102',20,19,7,47,93)

insert into Grade_report values('123456787','BT101',20,20,8,48,96)
insert into Grade_report values('123456787','BT102',20,18,8,46,92)

insert into Grade_report values('123456786','ECE101',20,18,7,48,93)
insert into Grade_report values('123456786','ECE102',19,19,7,45,90)

insert into Grade_report values('123456785','ISE101',20,19,7,37,83)
insert into Grade_report values('123456785','ISE102',20,19,7,38,84)

insert into Grade_report values('123456784','CSE101',20,19,9,47,95)
insert into Grade_report values('123456784','CSE102',20,19,7,47,93)
```

```sql
insert into Grade_report values('123456783','ISE101',20,20,8,48,96)
insert into Grade_report values('123456783','ISE102',20,18,8,46,92)

insert into Grade_report values('123456782','CSE101',20,18,7,48,93)
insert into Grade_report values('123456782','CSE102',19,19,7,45,90)

select * from Grade_report
```

	usn	course_id	mse1	mse2	task	see	total
1	123456782	CSE101	20	18	7	48	93
2	123456782	CSE102	19	19	7	45	90
3	123456783	ISE101	20	20	8	48	96
4	123456783	ISE102	20	18	8	46	92
5	123456784	CSE101	20	19	9	47	95
6	123456784	CSE102	20	19	7	47	93
7	123456785	ISE101	20	19	7	37	83
8	123456785	ISE102	20	19	7	38	84
9	123456786	ECE101	20	18	7	48	93
10	123456786	ECE102	19	19	7	45	90
11	123456787	BT101	20	20	8	48	96
12	123456787	BT102	20	18	8	46	92
13	123456788	CSE101	20	19	9	47	95
14	123456788	CSE102	20	19	7	47	93
15	123456789	ISE101	20	19	7	47	93
16	123456789	ISE102	20	19	7	48	94

CLASS table:

```sql
insert into Class values('123456789',5,'A')
insert into Class values('123456788',3,'B')
insert into Class values('123456787',7,'A')
insert into Class values('123456786',3,'A')
insert into Class values('123456785',5,'B')
insert into Class values('123456784',3,'B')
insert into Class values('123456783',5,'B')
insert into Class values('123456782',7,'B')

select * from Class
```

	usn	sem	sec
1	123456782	7	B
2	123456783	5	B
3	123456784	3	B
4	123456785	5	B
5	123456786	3	A
6	123456787	7	A
7	123456788	3	B
8	123456789	5	A

Write SQL queries to

1. List all the student details studying in fourth semester ‘B’ section.

   select * from Student

   SELECT S.USN, S.s_name, S.DOB, S.GENDER
   FROM CLASS C, STUDENT S
   WHERE C.USN=S.USN AND C.SEM=5 AND C.SEC='B'

2. Compute the total number of male and female students in each semester and in each section.

   SELECT C.SEM,C.SEC, COUNT (CASE WHEN S.GENDER='F' THEN 1 END) AS FEMALE_STUDENTS, COUNT (CASE WHEN GENDER='M' THEN 1 END) AS MALE_STUDENTS
   FROM STUDENT S, CLASS C

```
WHERE S.USN=C.USN
GROUP BY C.SEM, C.SEC
```

3. Create a view of MSE1 marks of student USN '4NM17IS001' in all subjects.

```
CREATE VIEW MSE1__MARKS AS
SELECT C.COURSE_ID, C.COURSE_TITLE, G.MSE1
FROM COURSE C, GRADE_REPORT G
WHERE C.COURSE_ID=G.COURSE_ID AND G.USN='123456782'

SELECT * FROM MSE1__MARKS
```

PART B

Create a database application using any programming language as the front end and any database software as a backend. For any problem selected, write the ER Diagram, apply ER-mapping rules, normalize the relations, and follow the application development process. Make sure that the application developed should have a minimum of five tables, at least one trigger and one stored procedure. Indicative areas include; health care, education, industry, transport, library.

COURSE OUTCOMES

At the end of the course the student will be able to

Sl. No.	Course Outcome (CO)	PI	Bloom's Taxonomy Level (BTL)
C508.1	Make use of ER diagrams concepts to design a database for a given real world scenarios.	1.3.1,1.4.1,2.4.4,4.2.1	**L3**
C508.2	Make use of Schema diagrams concepts to design a database for a given real world scenarios.	1.3.1,1.4.1,2.4.4,4.2.1	**L3**
C508.3	Analyze abstract problems and apply a combination of hardware and software to address problems. Implement database creation using Data Definition Language(DDL) concepts.	1.3.1,1.4.1,2.4.4,4.1.3,4.2.1, 5.1.2	**L4**
C508.4	Apply the DML(Data Manipulation Langauge) Concepts to query the Database	1.3.1,1.4.1,2.4.4,4.1.3,4.2.1, 5.1.2	**L3**
C508.5	Apply the concepts of complex queries in database environment	1.3.1,1.4.1,2.4.4,4.1.3,4.2.1, 5.1.2	**L3**

Mapping of POs & COs:

POs / Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
C508.1	1	2		2						1	1			
C508.2	1	2		2						1	1			
C508.3	2	2		2	2					1	1			3
C508.4	2	2		2	2					1	1			
C508.5	2	2		2	2					1	1			2

(L = 1: 30%-49%, M = 2: 50%-69%, H = 3: >70%)

TEXTBOOK:

1. Database Systems Models, Languages, Design and Application Programming, Ramez Elmasri and Shamkant B. Navathe, 7th Edition, 2017, Pearson.

REFERENCE BOOKS:

1. Database System Concepts- Silberschatz, Korth and Sudharshan, Sixth Edition, Mc-GrawHill, 2010

E-RESOURCES:

1. https://swayam.gov.in/nd1_noc19_cs46/preview , Database Management System, Swayam.
2. https://www.coursera.org/learn/intro-sql, Introduction to Structured Query Language (SQL), coursera.
3. https://www.coursera.org/projects/introduction-to-relational-database-and-sql, Introduction to Relational Database and SQL.

Note: In the examination, each student will pick one full question from 5 database scenarios.

****